

Program 10

Demonstrate Kubernetes Dashboard and CLI for Cluster Monitoring

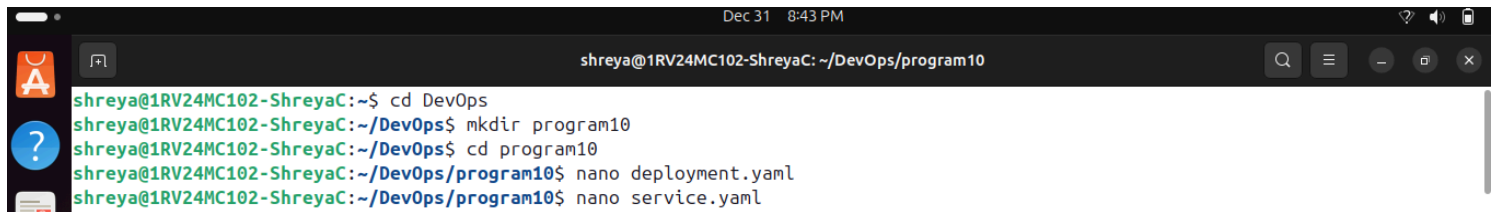
DevOps

— Program10

— deployment.yaml

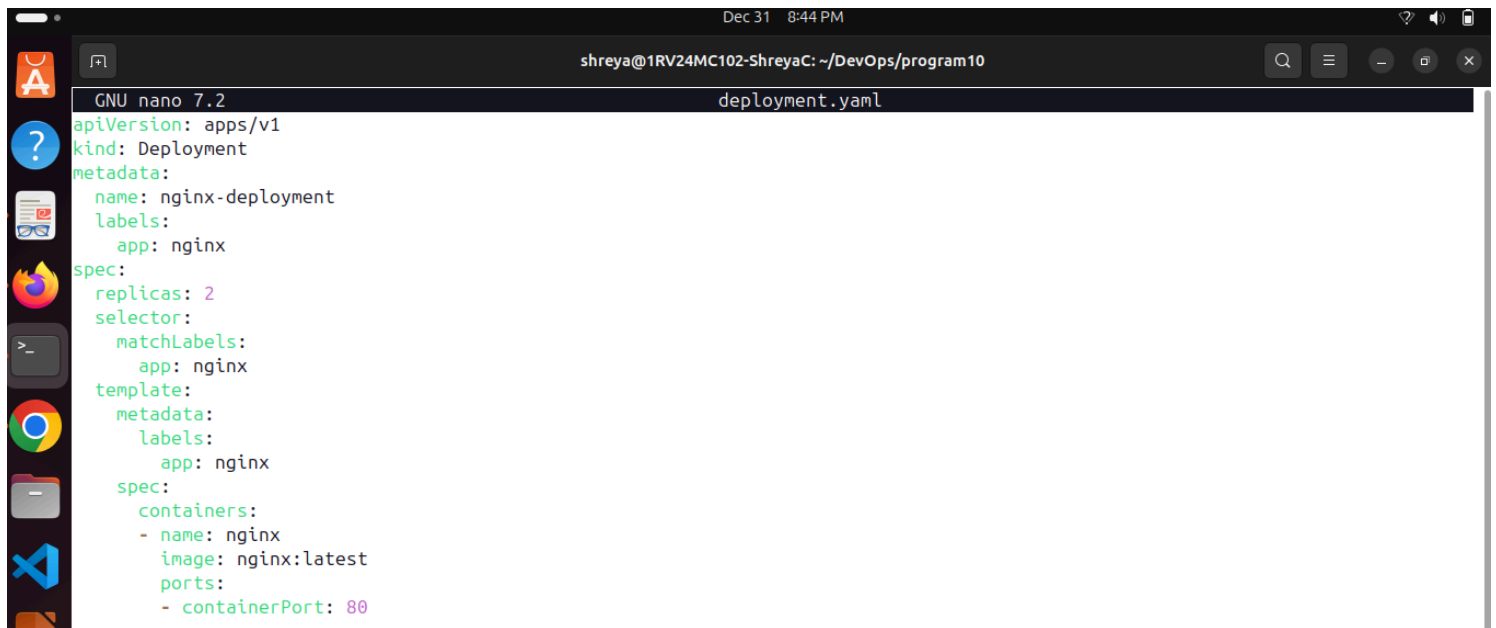
— service.yaml

Step 1 :



```
shreya@1RV24MC102-ShreyaC: ~$ cd DevOps
shreya@1RV24MC102-ShreyaC: ~/DevOps$ mkdir program10
shreya@1RV24MC102-ShreyaC: ~/DevOps$ cd program10
shreya@1RV24MC102-ShreyaC: ~/DevOps/program10$ nano deployment.yaml
shreya@1RV24MC102-ShreyaC: ~/DevOps/program10$ nano service.yaml
```

Step 2 : Add the following code in deployment.yaml file



```
GNU nano 7.2 deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
  labels:
    app: nginx
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80
```

Step 3 : Add the following code in service.yaml file

```
Dec 31 8:46 PM
shreya@1RV24MC102-ShreyaC: ~/DevOps/program10
GNU nano 7.2 service.yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
  - protocol: TCP
    port: 80
    targetPort: 80
  type: NodePort
```

Step 4 : Initiate minikube allocating specific CPUs and Memory

```
shreya@1RV24MC102-ShreyaC:~/DevOps/program10$ minikube start --cpus=2 --memory=4096
minikube v1.37.0 on Ubuntu 24.04
Using the docker driver based on existing profile
You cannot change the memory size for an existing minikube cluster. Please first delete the cluster.
Starting "minikube" primary control-plane node in "minikube" cluster
Pulling base image v0.0.48 ...
Restarting existing docker container for "minikube" ...
Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
Verifying Kubernetes components...
  ■ Using image registry.k8s.io/metrics-server/metrics-server:v0.8.0
  ■ Using image gcr.io/k8s-minikube/storage-provisioner:v5
Enabled addons: metrics-server, storage-provisioner, default-storageclass
Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
```

Step 5 : Enable Kubernetes Dashboard

```
shreya@1RV24MC102-ShreyaC:~/DevOps/program10$ minikube dashboard
Enabling dashboard ...
  ■ Using image docker.io/kubernetesui/dashboard:v2.7.0
  ■ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
Some dashboard features require the metrics-server addon. To enable all features please run:

    minikube addons enable metrics-server

Verifying dashboard health ...
Launching proxy ...
Verifying proxy health ...
```

Step 6 : Apply the resources for both the files

```
shreya@1RV24MC102-ShreyaC:~/DevOps/program10$ kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment created
shreya@1RV24MC102-ShreyaC:~/DevOps/program10$ kubectl apply -f service.yaml
service/nginx-service created
```

Step 7 : Get the pods

```
shreya@1RV24MC102-ShreyaC:~/DevOps/program10$ kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
demo-cron-29451860-sccrh	0/1	Completed	0	22h
demo-cron-29451862-sqcrm	0/1	Completed	0	22h
demo-cron-29453214-rcwbn	0/1	Completed	0	2m6s
demo-cron-29453216-fmpm2	0/1	ContainerCreating	0	104s
nginx-deployment-6f9664446b-5w9dt	0/1	ContainerCreating	0	15s
nginx-deployment-6f9664446b-ls2b5	0/1	ContainerCreating	0	15s
program-8	1/1	Running	1 (2m27s ago)	22h

Step 8 : Get the nodes

```
shreya@1RV24MC102-ShreyaC:~/DevOps/program10$ kubectl get nodes
NAME      STATUS    ROLES    AGE   VERSION
minikube  Ready     control-plane  36d   v1.34.0
```

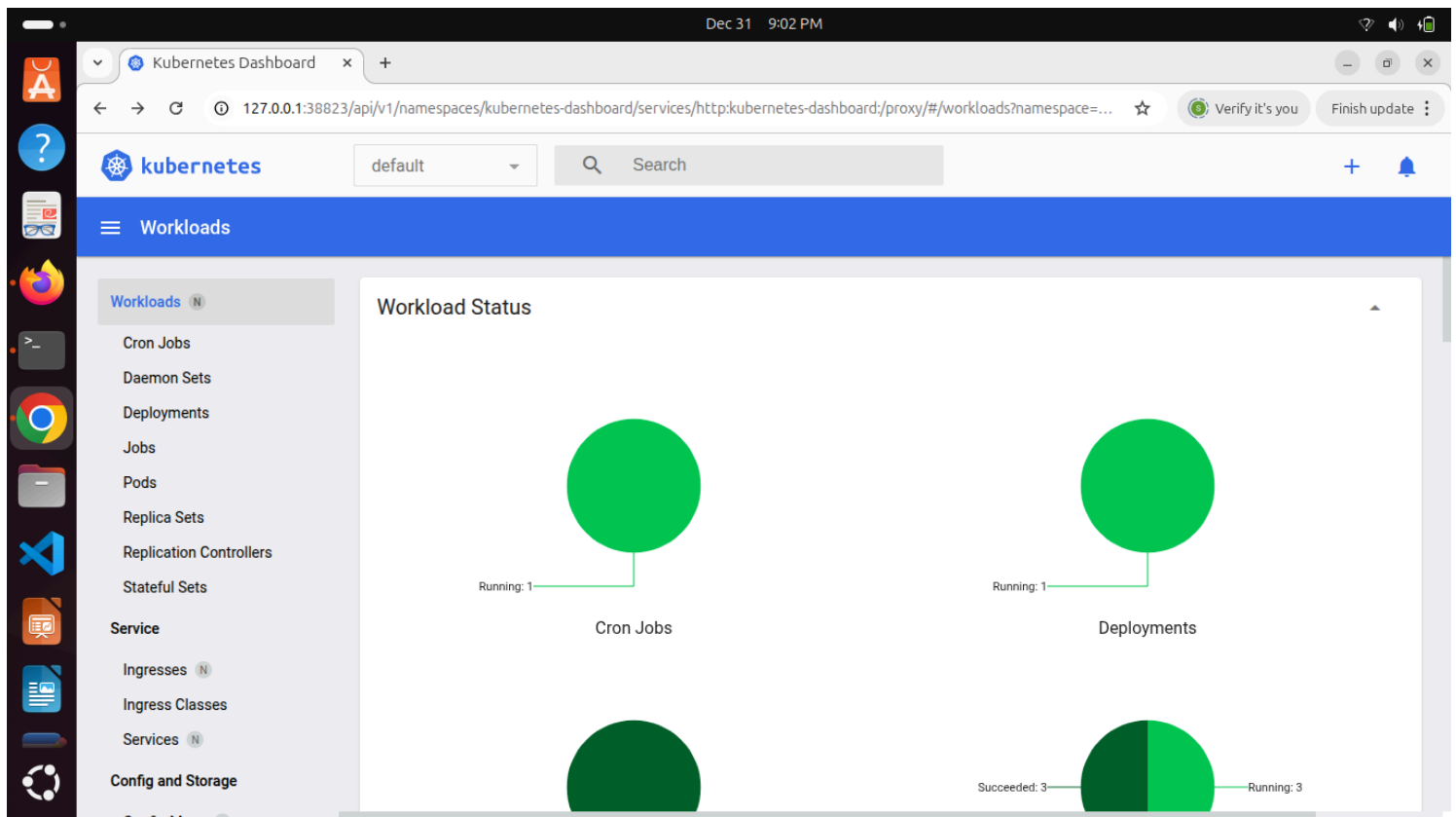
Step 9 : Check the pod logs

```
shreya@1RV24MC102-ShreyaC:~/DevOps/program10$ kubectl logs nginx-deployment-6f9664446b-5w9dt
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2025/12/31 14:57:59 [notice] 1#1: using the "epoll" event method
2025/12/31 14:57:59 [notice] 1#1: nginx/1.29.4
2025/12/31 14:57:59 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2025/12/31 14:57:59 [notice] 1#1: OS: Linux 6.14.0-36-generic
2025/12/31 14:57:59 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2025/12/31 14:57:59 [notice] 1#1: start worker processes
2025/12/31 14:57:59 [notice] 1#1: start worker process 29
2025/12/31 14:57:59 [notice] 1#1: start worker process 30
2025/12/31 14:57:59 [notice] 1#1: start worker process 31
2025/12/31 14:57:59 [notice] 1#1: start worker process 32
2025/12/31 14:57:59 [notice] 1#1: start worker process 33
2025/12/31 14:57:59 [notice] 1#1: start worker process 34
2025/12/31 14:57:59 [notice] 1#1: start worker process 35
2025/12/31 14:57:59 [notice] 1#1: start worker process 36
```

Step 10 : Describe the pods

```
shreya@1RV24MC102-ShreyaC:~/DevOps/program10$ kubectl describe pod nginx-deployment-6f9664446b-5w9dt
Name: nginx-deployment-6f9664446b-5w9dt
Namespace: default
Priority: 0
Service Account: default
Node: minikube/192.168.67.2
Start Time: Wed, 31 Dec 2025 20:27:29 +0530
Labels: app=nginx
pod-template-hash=6f9664446b
Annotations: <none>
Status: Running
IP: 10.244.1.27
IPs: IP: 10.244.1.27
Controlled By: ReplicaSet/nginx-deployment-6f9664446b
Containers:
  nginx:
    Container ID: docker://ddc7092c6714ac83276ae57dd8a5eb03da239cd6dd9387a077a0e38325db6e34
    Image: nginx:latest
    Image ID: docker-pullable://nginx@sha256:c8dc101ba61a499398c245fed8123abc20a3867ebd3664087890a054a781770b
    Port: 80/TCP
    Host Port: 0/TCP
    State: Running
      Started: Wed, 31 Dec 2025 20:27:58 +0530
    Ready: True
    Restart Count: 0
    Environment: <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-m78l7 (ro)
Conditions:
  Type Status
```

Step 11 : Access the application



Step 12 : Clean up all the process

```
shreya@1RV24MC102-ShreyaC:~/DevOps/program10$ kubectl delete pods --all
pod "demo-cron-29453248-jrfl1" deleted from default namespace
pod "demo-cron-29453250-gn5mn" deleted from default namespace
pod "demo-cron-29453252-d8rwr" deleted from default namespace
pod "nginx-deployment-6f9664446b-5w9dt" deleted from default namespace
pod "nginx-deployment-6f9664446b-ls2b5" deleted from default namespace
pod "program-8" deleted from default namespace
```